

The allegory axioms, and what defines them in the Frobenius calculus

§1. Cartesian bicategory of relations

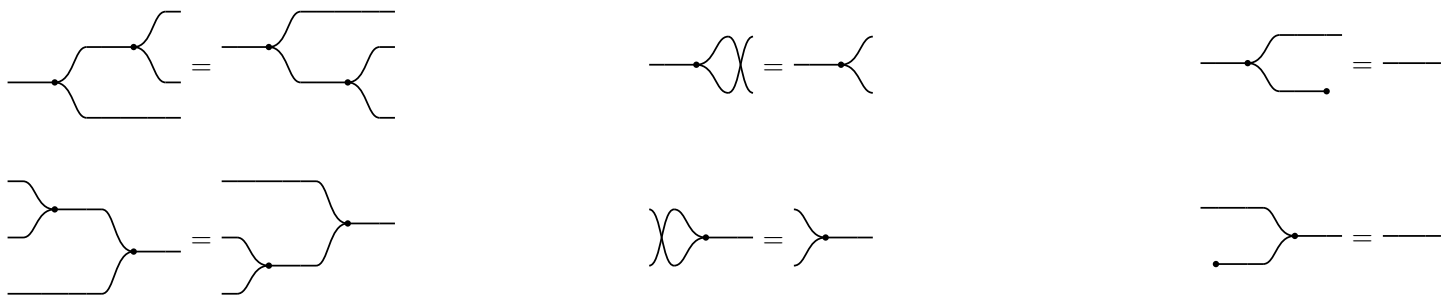
Definition 1.1.

A **cartesian bicategory of relations** is a poset-enriched category(bicategory) that is has a symmetric monoidal product $\otimes$ (cartisn product of set in Rel, vertical junctposition in the diagram) and

$\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$, **Frobenius**, arrows **lax**.

$(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow \mathbb{I})$ formed a cocommutative comonoid

$(\triangleright : A \otimes A \rightarrow A, \multimap : \mathbb{I} \rightarrow A)$ formed a commutative monoid



In $\text{Rel}(\text{Set})$, the four generators are these relations on a set A :

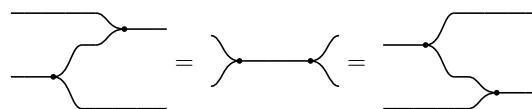
- $\triangleleft = \{(a, (a, a))\}$ — copy: one in, two identical out
- $\triangleright = \{((a, a), a)\}$ — merge: a pair passes only when its two components agree, and the common value comes out
- $\multimap = A \times \{*\}$ — discard: forget the value, keep only that there was one
- $\multimap = \{*\} \times A$ — create: any element at all

§1.1. $\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$



The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In Rel it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§1.2. The Frobenius law



The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, $\triangleright \triangleleft$ — merge, then copy — which is the two-in two-out **spider**. Every connected diagram built from these four generators collapses to the spider on its own inputs and outputs, and this equation is what starts that collapse. A **bubble** is the other order, $\triangleleft \triangleright$: copy then merge, a closed loop, which the idempotency of n uses below.

§1.3. Every arrow is a lax comonoid homomorphism

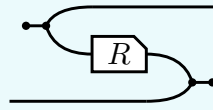


$$R\triangleleft \leq \triangleleft(R\otimes R)$$

§2. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

Definition 2.1.

The **converse** R° of $R : a \rightarrow b$ is R with both of its wires turned round,

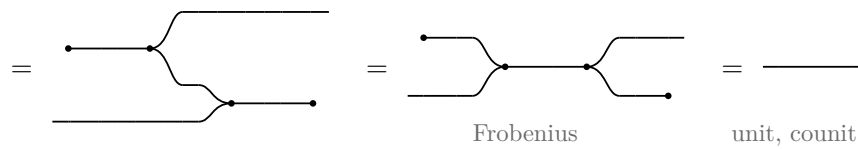


$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

where $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. And it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

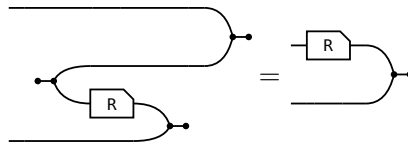
$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§2.1. $1^\circ = 1$ (snake)



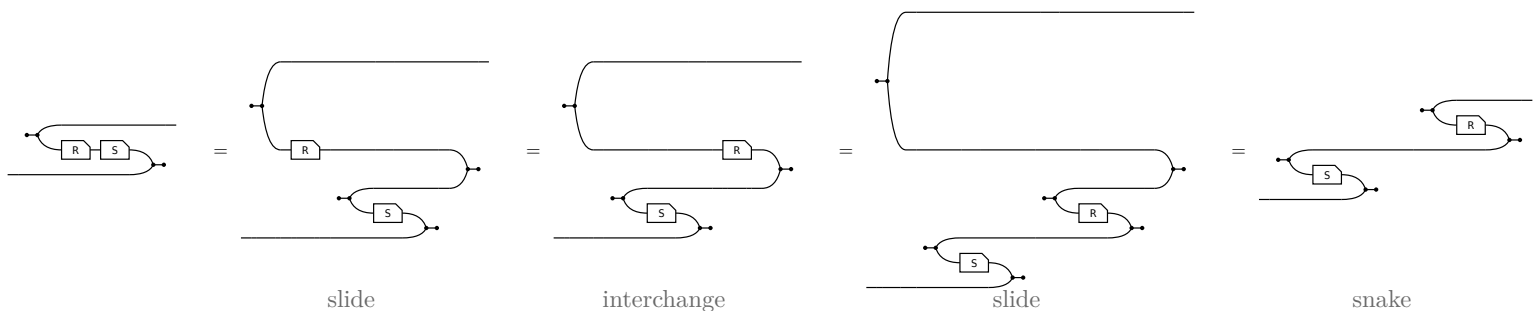
§2.2. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



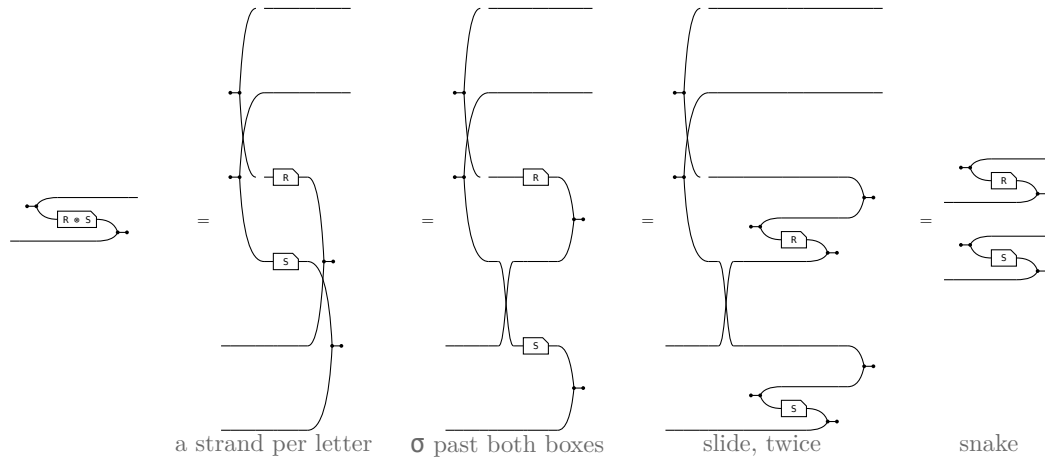
R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circlearrowright$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circlearrowleft \triangleleft$ bends the a strand down and the $\triangleright \circlearrowright$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

§2.3. $(R \ S)^\circ = S^\circ R^\circ$



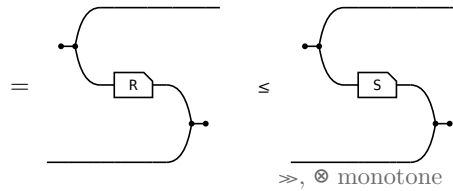
§2.4. $(R \otimes S)^\circ = R^\circ \otimes S^\circ$

A merge at a product is two merges behind a crossing, so the pair unbends one strand at a time and the crossing is all that is left to move.



§2.5. $R \leq S$ implies $R^\circ \leq S^\circ$

R sits in a frame of wires built from $\gg$ and $\otimes$, and both of those are monotone, so the box may be replaced where it stands.



§3. Every arrow is a lax monoid homomorphism

The merge/create mirror of the two comonoid laws opening this note — the paper's (41) and (42), p. 22, which it gets by bending them with (ii) and (iv) above. No bending is needed: expand by the unit $1 \leq \triangleleft \triangleright$, duplicate the box with the comonoid law, contract by the counit $\triangleright \triangleleft \leq 1$. Both halves of the adjunction, and nothing else.

In $\mathbf{Rel}(\mathbf{Set})$ with $R \subseteq A \times B$, where $\triangleright$ is the equality test on a pair and $\bullet$ produces any element at all:

$$\begin{aligned}
 \triangleright R &= \{((a, a), b) : a R b\} \text{ — two equal inputs, then one run of } R \\
 (R \otimes R) \triangleright &= \{((a, a'), b) : a R b \text{ and } a' R b\} \text{ — a run on each input, then the results merged} \\
 \bullet R &= \{(*, b) : \exists a. a R b\} \text{ — the image of } R \\
 \bullet &= \{*\} \times B \text{ — all of } B
 \end{aligned}$$

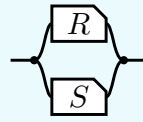
Take $a' = a$ for the first containment; every b lies in B for the second. Each is strict exactly when its reverse fails — R not injective, R not surjective — which is what the later table on maps reads off them.

inequation, and what proves it	picture
$(41) \triangleright R \leq (R \otimes R) \triangleright$ $\triangleright R = \triangleright R 1 \leq \triangleright R \triangleleft \triangleright \leq \triangleright \triangleleft (R \otimes R) \triangleright \leq (R \otimes R) \triangleright$ — unit, lax $\triangleleft$, counit.	
$(42) \bullet R \leq \bullet$ $\bullet R = \bullet R 1 \leq \bullet R \bullet \bullet \leq \bullet \bullet \bullet \leq \bullet$ — the same three steps at $\mathbb{I}$.	

§4. $\cap$ is a commutative idempotent monoid on every hom-set

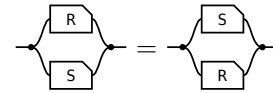
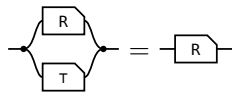
Definition 4.1.

The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



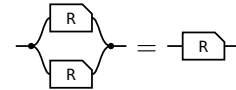
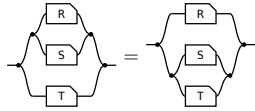
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).



unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$

commutative: σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

semi-distributivity, and what supplies it	picture
$R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F (R \cap S) = F R \cap F S$.	

§5. Unitary and pre-tabular — what the other direction needs

condition	allegory	Frobenius side
unitary	τ is a unit when 1τ is the largest endomorphism on it and every object carries an entire arrow to it. A unit is what makes τ exist.	free. I is the unit object, $1 \leq \multimap \multimap$ says $\multimap$ is entire, and $\tau = \multimap \multimap$ is maximal.
pre-tabular	f, g tabulate R when both are maps, $R = f \circ g$, and R is tabular when some pair tabulates it, the allegory pre-tabular when every arrow lies under a tabular one.	this is what $\otimes$ is. Given unitarity it reduces to tabulating each τ , and a tabulation of τ is an object $n \otimes m$ with its two projections. One side derives the product from tabulations, the other posits it.

Together: **cartesian bicategory of relations** $\simeq$ **unitary pre-tabular allegory**.

§6. Union

Past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure. It draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. The residual, which needs a second composition with linear adjoints, is in the division section below.

statement	picture
$R \sqsubseteq R \cup S$ union $R S := \langle R, S \rangle [1, 1]$: offer both branches, then forget which was taken.	
$R \cup S = S \cup R$	
$\perp \cup R = R$ $\perp$ routes through the zero object; there is no shape for it.	
$R (S \cup T) = R S \cup R T$ — composition distributes over union, which $\cap$ does not.	
$(R \cup S)^\circ = R^\circ \cup S^\circ$	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ — what makes the whole distributive , and the one picture with both layers in it.	

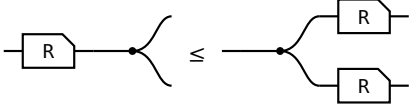
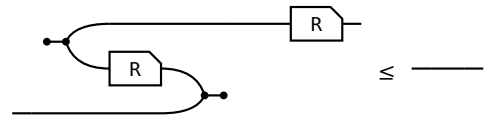
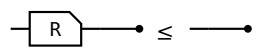
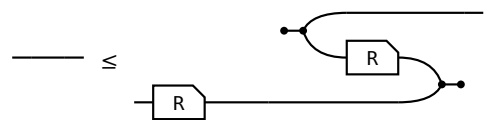
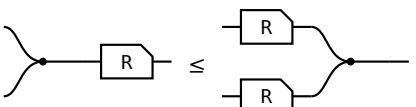
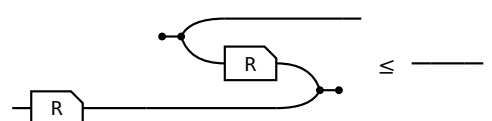
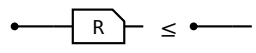
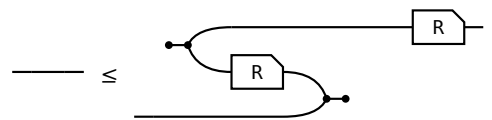
The last of Freyd's §2.21 equations that have no picture here — a tape around nothing, or a tape around what is already inside it, is not worth drawing. R = shops with rice, S = shops with sugar throughout.

equation	the rice reading
$R \cup (S \cap R) = R$ $(R \cup S) \cap R = R$ absorption	Shops with rice, or with both: still the shops with rice. Shops with rice or sugar, and with rice: the same. The smaller side of each pair is already inside the larger.

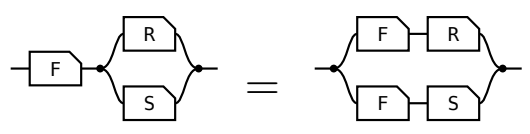
equation	the rice reading
$R \perp = \perp R$ $0_S = 0_{\{R S\}}$ — and $\perp R = \perp$ on the other side	Going to a shop and then taking a road that leads nowhere gets you nowhere. $\perp$ is a two-sided zero for composition, as τ is not.

§7. Maps

Every arrow is lax for $\triangleleft$ and for $\multimap$ by axiom, and for $\triangleright$ and $\multimap$ by the lax monoid section. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

holds for every arrow	property	holds iff
$R \triangleleft \leq \triangleleft (R \otimes R)$ 	(SV) single valued $R^\circ R \sqsubseteq 1$	
$R \multimap \leq \multimap$ 	(TOT) total $1 \sqsubseteq R^\circ R$. With single valued, a map.	
$\triangleright R \leq (R \otimes R) \triangleright$ 	(INJ) injective $R R^\circ \sqsubseteq 1$	
$\multimap R \leq \multimap$ 	(SUR) surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	

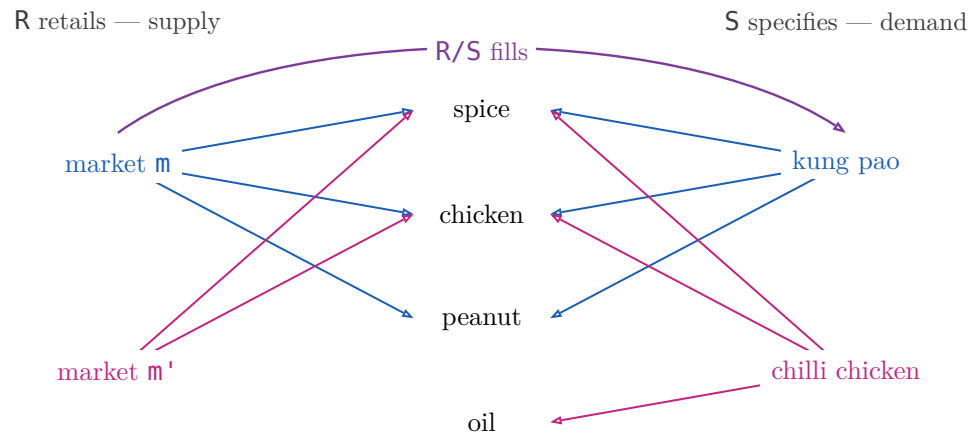
What (SV) buys, in one law. Only single valuedness is spent: F may be partial, and entirety is never used.

law	picture
$F (R \cap S) = F R \cap F S$ F single valued F = shops Ann may go to, R = has rice, S = has sugar. Single valued means one shop, so the sides agree.	 one shop with both rice at A, sugar at B

§8. Division is WHAT!

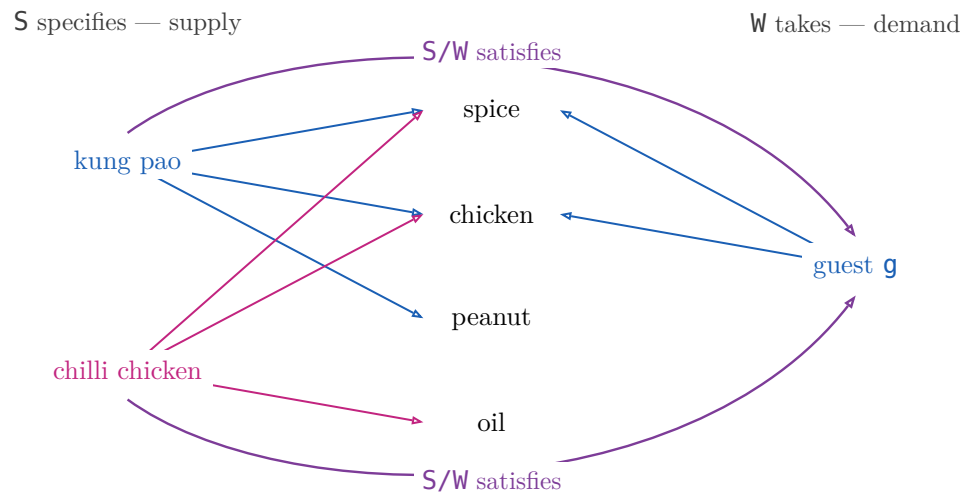
$$m \text{ (R/S) } d \iff \forall i. d \text{ S } i \rightarrow m \text{ R } i$$

Read the bar as **what**: R/S relates a producer to a consumer when the producer supplies *what* the consumer asks for. Each picture below is one such reading — supply on the left, demand on the right, the ingredients they meet over in the middle.



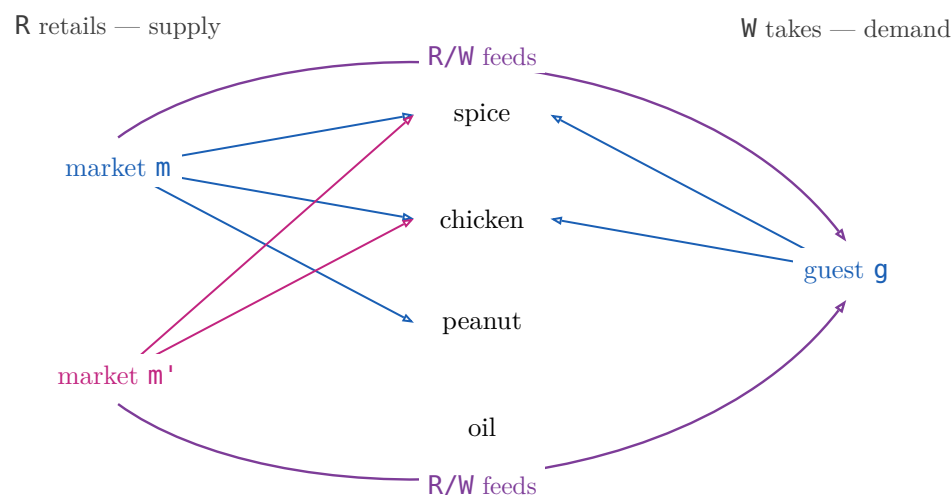
R/S — market retails *what* dish specifies

Every arrow the dish sends into the ingredient column has to be matched by one from the market, so here a single pair survives: m' fills no dish, lacking both peanut and oil, and chilli chicken is filled by no market, since neither stocks oil.



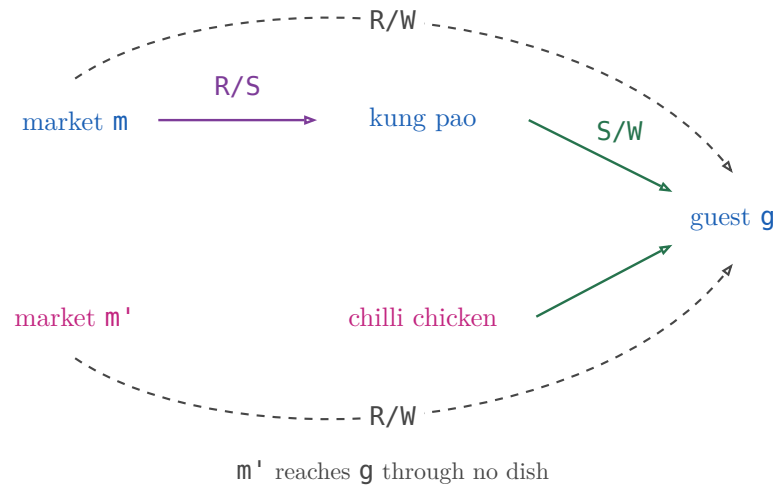
S/W — dish specifies *what* guest takes

The same test with the guest as the consumer. g takes spice and chicken, and both dishes need both, so both satisfy him — a dish may need more, never less.



R/W — market retails *what* guest takes

And with the market as the producer: both markets sell spice and chicken, so both feed g — m' included, though it fills no dish. Dropping the ingredients now, the three arcs sit like this:



$(R/S)(S/W)$ is a path: $m \rightarrow \text{kung pao} \rightarrow g$, and that is all of it. R/W also holds of m' , which retails everything g wants — but m' fills no dish, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

$X \sqsubseteq R/S \iff X S \sqsubseteq R$ X is any market-to-dish pairing; one that only pairs a market with dishes it fills lies inside R/S , and R/S is the largest such.	
$X \sqsubseteq S \setminus R \iff S X \sqsubseteq R$ The mirror — divide on the left when the market comes first.	
$(R/S) S \sqsubseteq R$ There is a dish m fills that specifies i — then m retails i too. Strict at $S = \emptyset$: R/S is everything, $(R/S) S = \emptyset$.	
$S (S \setminus R) \sqsubseteq R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A dish of dishes: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f (R/S) = (f R)/S$ Rename the market before or after dividing — the licence to write $f R / S$.	
$R/(f S) = (R/S) f^\circ$ Rename the dish: a map leaves a denominator as f° outside the box.	
$(R/S)(S/W) \sqsubseteq R/W$ A market that fills a dish that satisfies a guest feeds that guest.	
$1 \sqsubseteq R/R$ R/R runs market to market: each fills its own list. Strict: two markets retailing only spice fill each other's and stay two markets.	
$(R/R)(R/R) = R/R$ R/R is the preorder retails at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	

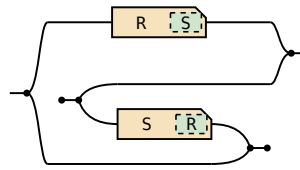
$(R/R) \ R = R$ Reaching $\dot{\mathbf{i}}$ through a market whose list $\mathbf{m}$ fills is reaching it directly, since $\mathbf{m}$ fills its own.	
$R/1 = R$ Filling the dish that specifies exactly $\dot{\mathbf{i}}$ is retailing $\dot{\mathbf{i}}$.	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Filling a dish made of two is filling each of them.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: $\cap$, $\cup$, $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§9. Symmetric division

$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ$. In $\mathbf{Rel}$ it relates $\mathbf{m}$ and $\mathbf{d}$ when they reach the same things, $\forall i$. $(\mathbf{m} \ R \ i \iff \mathbf{d} \ S \ i)$ — market $\mathbf{m}$ retails exactly what dish $\mathbf{d}$ specifies, no more and no less.

A meet of two long divisions, the second turned round by the converse frame:



exported from the definition, not transcribed

$\left(\frac{R}{S}\right)^\circ = \frac{S}{R}$ Matching is symmetric.	
$\frac{R}{S} \frac{S}{W} \sqsubseteq \frac{R}{W}$ And transitive.	

No pictures for the rest of §2.35: symmetric division is not built from the generators.

law	the market reading
$X \sqsubseteq \frac{R}{S} \iff XS \sqsubseteq R$ and $X^\circ \ R \sqsubseteq S$	X may pair $\mathbf{m}$ with $\mathbf{d}$ only when each fills the other. Both halves must typecheck, so the operation is partial .
$\frac{R}{S} S \sqsubseteq R$	$\mathbf{m}$ matches $\mathbf{d}$ and $\mathbf{d}$ specifies $\mathbf{i}$, so $\mathbf{m}$ retails $\mathbf{i}$.
$\frac{R}{R} R \sqsubseteq R$	The same with R against itself, market to market.
$1 \sqsubseteq \frac{R}{R}$	Every market matches itself.
$\left(\frac{R}{R}\right)^2 = \frac{R}{R}$	So matches is an equivalence relation. Freyd writes $\sqsubseteq$; with the row above it is an equality.
$X \sqsubseteq \frac{R}{R} \iff XR \sqsubseteq R$, for symmetric X	The largest symmetric arrow that leaves R alone.
$\frac{S}{S} = 1$, S is straight	No two dishes specify the same ingredients. Equivalently every symmetric X with $X \ S \sqsubseteq S$ is coreflexive.
$\mathbf{f} \ S = \mathbf{g} \ S \implies \mathbf{f} = \mathbf{g}$, S straight	A straight S tells its dishes apart, so it cancels on the right.

law	the market reading
$S \circ R \text{ straight} \implies S \text{ straight}$	If the longer chain tells them apart, the first step already does.
$S \text{ straight} \iff R/S \text{ simple for all } R$	If no two dishes agree, at most one market can match a given dish.
$R = h \circ S$, h a cover, S straight	In an effective division allegory every arrow factors that way.
$\frac{R}{1}$ is the simple part of R	The markets retailing one ingredient and nothing else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{Dom } \frac{R}{S} = 1 \cap (R/S) (S/R)$	Its domain is the domain of simplicity of R .

§10. Power allegories

One more operation on a division allegory: $\exists$, Freyd's **epsilon**. In $\mathbf{Rel}$ its source is the powerset of its target and $\mathfrak{l} \exists \mathfrak{i}$ iff $\mathfrak{i} \in \mathfrak{l}$ — the shopping lists over the ingredients, and $\exists$ reads a list $\mathfrak{l}$ back into the ingredients on it. Everything else is forced.

law	the market reading
$\exists \circ \tau = R \circ \tau$	Same target as R , and depending on nothing else: one $\exists$ per object, not per arrow. What is being listed fixes it, not who lists it.
$\frac{\exists}{\exists} \sqsubseteq 1$, $\exists$ is straight	Extensionality : two lists with the same ingredients are the same list.
$1 \sqsubseteq (R/\exists) (\exists/R)$, $\exists$ is thick	Comprehension : every market has a list of exactly what it retails. Equivalently every R factors as a map followed by $\exists$.
$\Lambda(R) \triangleq \frac{R}{\exists}$	The list-of map: send a market to its list.
$\Lambda(R)$ is simple	$\exists$ is straight, and dividing by a straight arrow is simple. At most one list per market.
$\Lambda(R)$ is entire $\iff \exists$ thick	$\text{Dom } \frac{R}{\exists} = 1 \cap (R/\exists) (\exists/R)$, the domain row above. At least one list per market, so $\Lambda(R)$ is a map .
$\Lambda(R) \circ \exists = R$	Look up a market's list, then read off its ingredients: what it retails.
$\Lambda(R)$ is the only map with $\Lambda(R) \circ \exists = R$	Two maps naming the same ingredients name the same list — extensionality again.
$F \sqsubseteq \Lambda(F \circ \exists)$, F simple	A partial choice of lists is inside the total one.
$[\alpha] = \text{source of } \exists$, the power-object	Every list over α .
$\{\cdot\} \triangleq \Lambda(1)$, the singleton map , monic	The one-ingredient list. $\Lambda(1) \circ \Lambda^\circ(1) \sqsubseteq \frac{1}{\exists} \circ \frac{1}{1} \sqsubseteq \frac{1}{1} \sqsubseteq 1$.
$\Lambda(f) = f \circ \{\cdot\}$, f a map	Rename first or take singletons first.
$\mathcal{C}$ a topos $\implies \mathbf{Rel}(\mathcal{C})$ a power allegory	And back: a unitary tabular power allegory has $\mathbf{Map}(\mathcal{A})$ a topos. Extensionality and comprehension are all a topos adds.